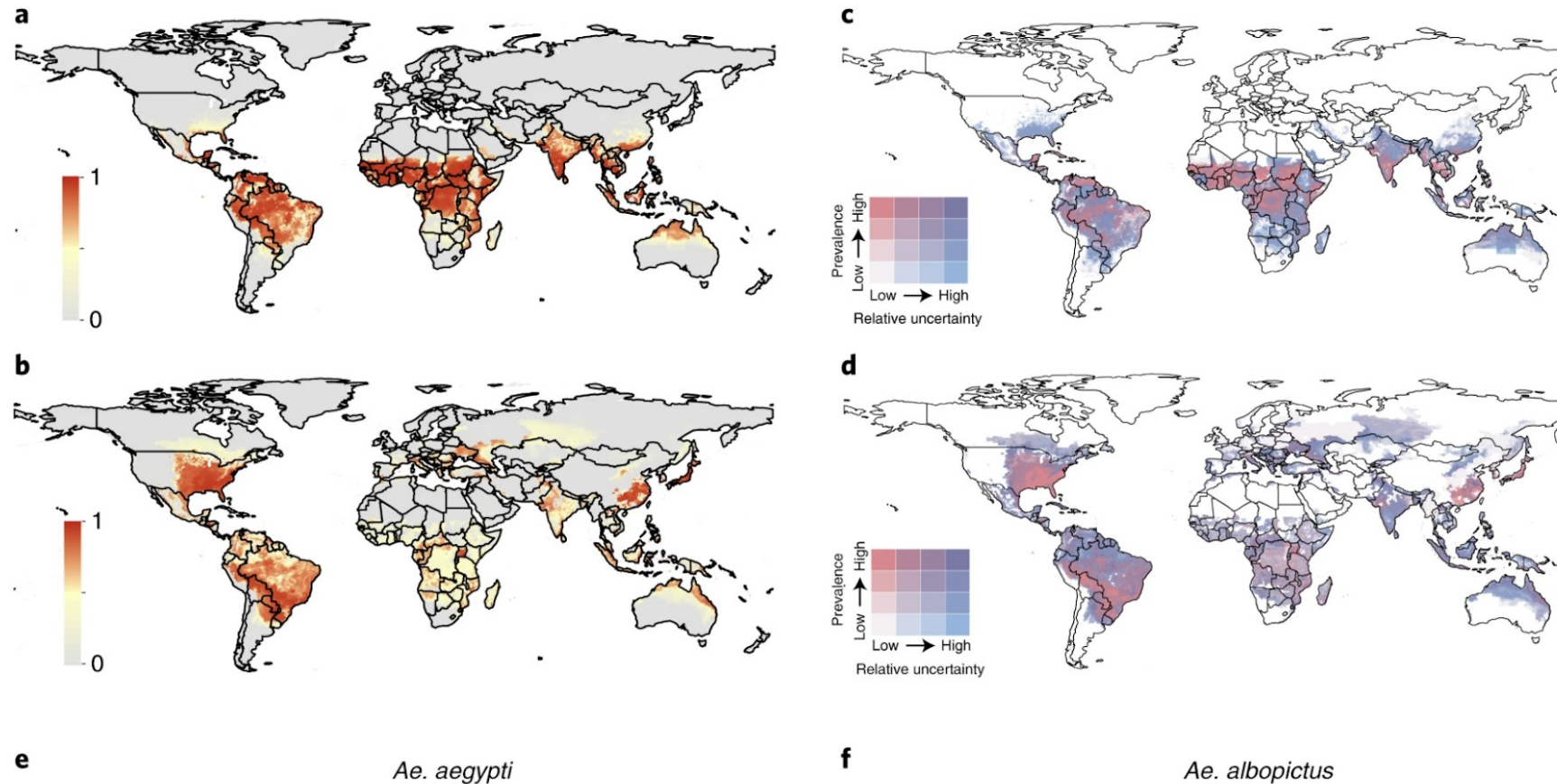


Climate-Sensitive Vector Dynamics
Modelling Workshop
September 17-19, 2025 - Nicosia, Cyprus



Summary of open access observational vector presence and abundance datasets

william.wint@gmail.com

DATA NEEDS



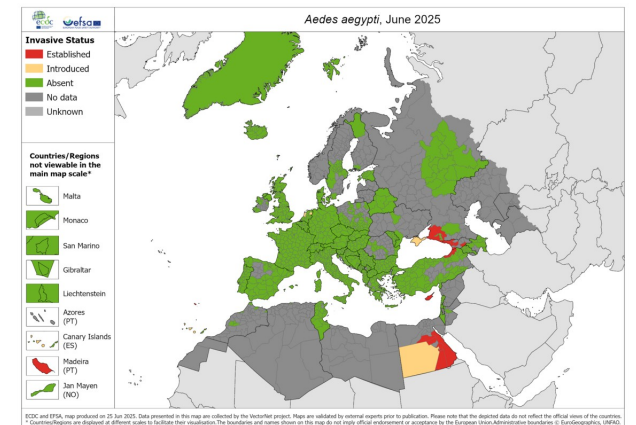
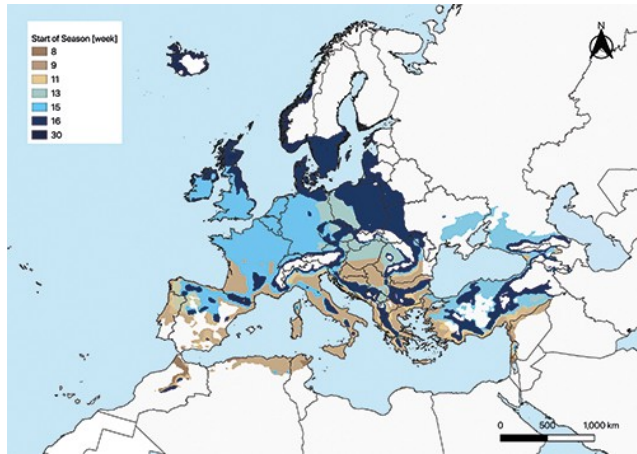
What are *Aedes* data needed for:

Mechanistic population models (validation)
based on abundance, life stage, traits etc

Spatial distribution models of
presence/absence
abundance
NB also life stage/sex

Seasonal activity
season start
season end
season length

Dissemination and illustration, advocacy
simplified maps, overviews



Variables:

DATA VARIABLES



Presence (and absence)

Useful for

seasonality and activity

dependent on complete sampling

Less useful for mechanistic/deterministic models

Absence

Just as important, at least for distribution modelling

Rarely provided

Often need to generate

Seasonality

Start end dates, peak numbers/sample effort

Abundance

?? Related to risk??

How to measure?

? Complete season needed

Standardised indices/Max number

Standardised collection methods

How to equilibrate counts of different life stages

SOURCES: FIELDWORK



Usually snapshots and limited spatial and temporal coverage,
may be ok for deterministic modelling as often site specific
needs to be aggregated for use at scale
standard metrics essential

VectorNet: **will become more relevant in coming years**

field work over six years, targeted to fill distribution gaps
standardised sampling strategy
20+ mosquito species
historically pres abs , recently abundance with effort



AIMSURV – voluntary - EU **albopictus focused**, designed for abundance
modelling

42 teams, 24 countries
standardised ovitrap sampling methodology, two tier sampling strategy
Template for aegypti in future

Many many smaller scale reported in literature (vectorByte) - **global**

SOURCES: LITERATURE



Literature

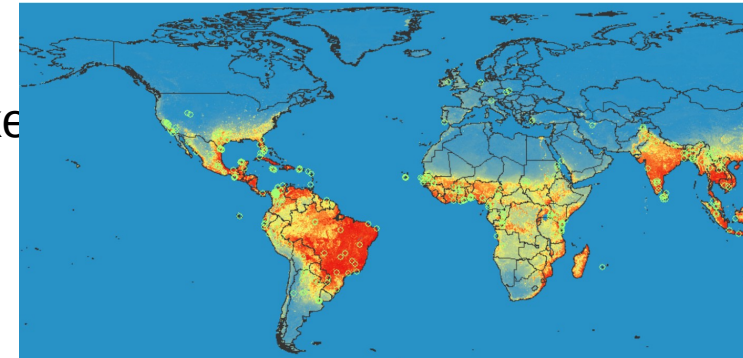
available but very labour intensive
standardisation and extraction an issue – also publications may be incomplete

Kraemer *et al* remains major example but is now old, presence only

e4warning informal update

publish or perish – quicker

programmatic scanning,
aegypti
presence only
easy to make errors

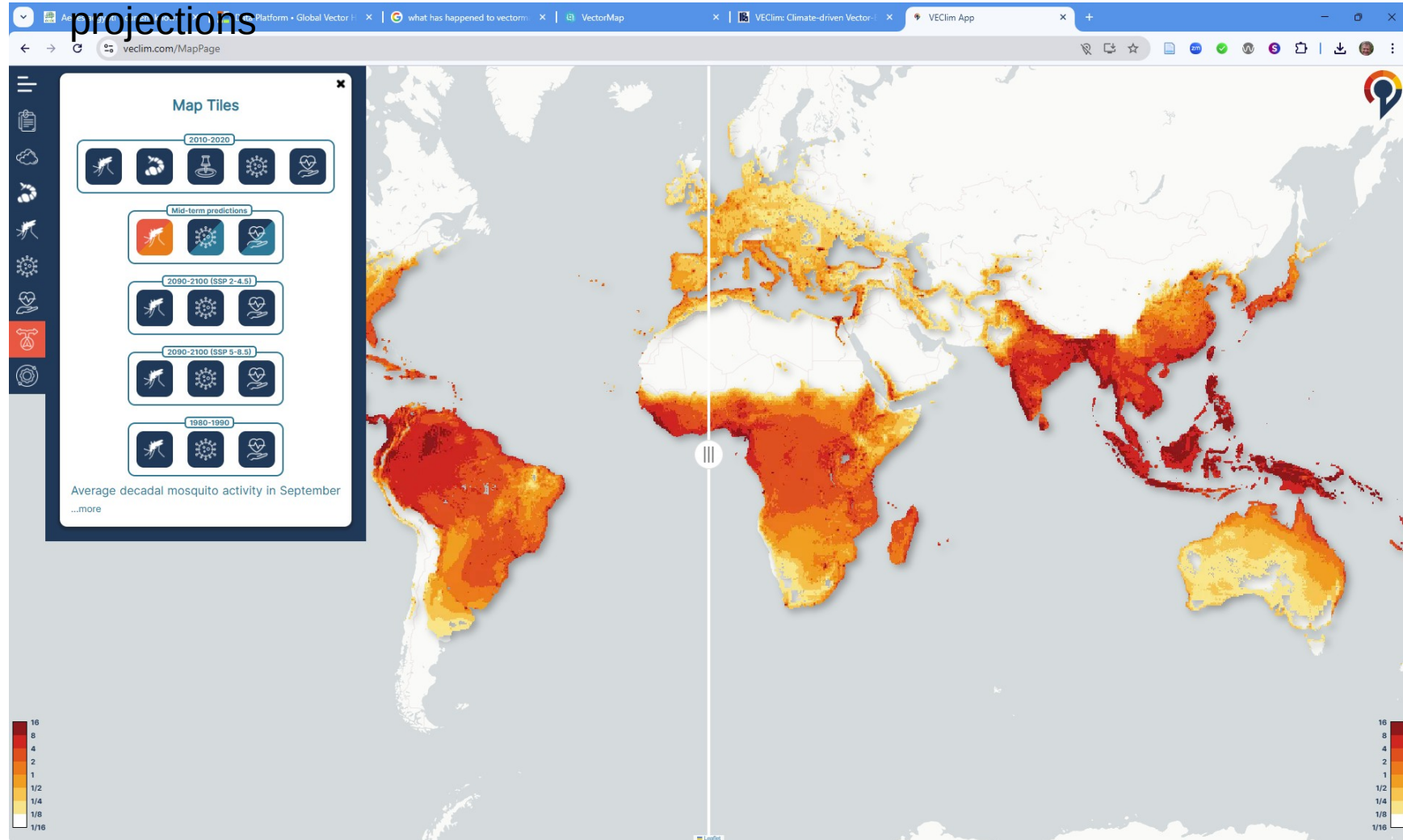


abundance data much more difficult and no comprehensive attempt yet made

SOURCES: easy to get at, more difficult to process especially for abundance ONLINE



VEClim : albopictus/ detailed lifestage and risk forecasts and
projections



William Wint, Available *Aedes* datasets, Modelling workshop, Cyprus, Sept 25

SOURCES:

Vectorbyte.org traits (40 aegypti, 50 albo) and abundance database lists



The screenshot shows a web browser with multiple tabs open, including 'Aedes aegypti - current know...', 'Data Platform - Global Vector...', 'what has happened to vectorm...', 'VectorMap', and 'Vector Data Ecosystem'. The address bar shows 'vectorbyte.org/vecdataecosystem'. The website has a navigation bar with links: Home, Our Team, FAQs, Training, Blog, Outputs, Databases, and a 'Sign In/Sign Up' button. The main content area features the title 'Characterizing the Vector Data Ecosystem' and a paragraph explaining the growing body of information on vector-borne diseases. Below this, there is a section for 'Global Biological Information Facility (GBIF) depositors and aggregators' with a table listing GBIF, GBIF.us (Formerly BISON), and SCAN. A cookie consent banner is visible at the bottom.

Characterizing the Vector Data Ecosystem

A growing body of information on vector-borne diseases has arisen as increasing research focus has been directed towards the need for anticipating risk, optimizing surveillance, and understanding the fundamental biology of vector-borne diseases to direct efforts to control and mitigation. The scope and scale of this information, in the form of data, comprising database efforts, data storage, and serving approaches, mean that it is distributed across many formats and data types. Data ranges from collections records to molecular characterization, geospatial data to interactions of vectors and traits, infection experiments to field trials. New initiatives arise, often spanning the effort traditionally siloed in specific research disciplines, and other efforts wane, perhaps in response to funding declines, different research directions, or lack of sustained interest. Thusly, the world of vector data – the Vector Data Ecosystem – can become unclear in scope, and the flows of data through these various efforts can become stymied by obsolescence, or simply by gaps in access and interoperability. As increasing attention is paid to creating FAIR (Findable Accessible Interoperable, and Reusable) data, simply characterizing what is 'out there', and how these existing data aggregation and collection efforts interact, or interoperate with each other, is a useful exercise. This website and related project presents a snapshot of current vector data efforts, a brief description of their stated scope and purpose, and level of accessibility. We [welcome additions](#) to this set, following a similar template, to improve this resource for the larger vector ecology and vector-borne disease research and practitioner community.

We invite you to [view our blog post](#) about this work and see the [publication](#) on the database which includes more context and commentary.

(FO) Fully Open Database
(PA) Partially Accessible Database

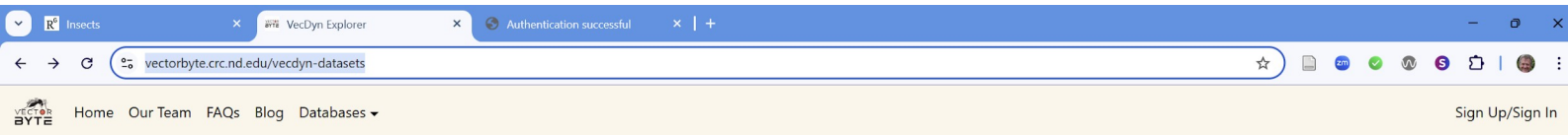
Global Biological Information Facility (GBIF) depositors and aggregators

GBIF	GBIF.us (Formerly BISON)	SCAN
------	--------------------------	------

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SOURCES:

VectorByte VecDyn App individual collections of abundance datasets,
<https://vectorbyte.crc.nd.edu/vecdyn-datasets>



Instructions +

Search by phrase...

Select a columnSelect an operatorEnter a search term...

SearchClearDownload

Click any row in the table to view its details page or click the checkbox beside any number of rows and then click the download button to download their data.

Select	Species Name	Title	Co
☐	Aedes aberratus, Aedes aurifer, Aedes canadensis complex, Aedes cinereus, Aedes grossbecki, Aedes japonicus complex, Aedes nigromaculis, Aedes sollicitans, Aedes sp., Aedes sticticus, Aedes thibaulti, Aedes triseriatus, Aedes trivittatus, Aedes vexans, Anopheles barberi, Anopheles crucians, Anopheles punctipennis, Anopheles quadrimaculatus, Anopheles sp., Anopheles walkeri, BLANK, Coquillettidia perturbans, Culex pipiens morphological group, Culex salinarius, Culex sp., Culex tarsalis, Culex territans, Culicidae, Culiseta inornata, Culiseta melanura, Culiseta sp., Psorophora ciliata, Psorophora cyanescens, Psorophora ferox, Psorophora howardii, Psorophora sp., Uranotaenia sapphirina	2023 Mosquito Surveillance St Joseph County Indiana USA	208
☐	Aedes cinereus, Aedes dorsalis, Aedes fitchii, Aedes increpitus, Aedes melanimon, Aedes nigromaculis, Aedes sierrensis, Aedes sticticus, Aedes vexans, Anopheles freeborni, BLANK, Coquillettidia perturbans, Culex pipiens, Culex tarsalis, Culiseta incidens, Culiseta inornata, Culiseta minnesotae	2021-2023 Ada, Idaho	524
☐	Aedes aegypti, Aedes albopictus, Aedes atlanticus, Aedes infirmatus, Aedes sp., Aedes taeniorhynchus, Aedes vexans, Anopheles bradleyi, Anopheles crucians complex, Anopheles punctipennis, Anopheles quadrimaculatus, Anopheles sp., BLANK, Coquillettidia perturbans, Culex erraticus, Culex nigripalpus, Culex quinquefasciatus, Culex sp., Culiseta melanura, Mansonia genus sp., Mansonia titillans, Melanoconion Sp., Psorophora ciliata, Psorophora columbiae, Psorophora ferox, Psorophora sp., Toxorhynchites rutilus, Unknown unknown, Uranotaenia lowii, Uranotaenia sapphirina, Uranotaenia sp., Wyeomyia sp.	2010-2019 Collier County	406
☐	Aedes albopictus, Aedes canadensis complex, Aedes cinereus, Aedes hendersoni, Aedes japonicus complex, Aedes sollicitans, Aedes sticticus, Aedes stimulans, Aedes triseriatus, Aedes trivittatus, Aedes vexans, Anopheles barberi, Anopheles punctipennis, Anopheles	2021-2023 Mosquito Surveillance, Desplaines Valley Mosquito Abatement District (Chicago-land)	412

VecDyn Dataset: 218

vectorbyte.crc.nd.edu/vecdyn-detail/218

Home Our Team FAQs Blog Databases

Sign Up/Sign In

Experimental Information

Dataset Number: 218

Species: Aedes aegypti, Aedes albopictus, Aedes atlanticus, Aedes infirmatus, Aedes sp., Aedes taeniorhynchus, Aedes vexans, Anopheles bradleyi, Anopheles crucians complex, Anopheles punctipennis, Anopheles quadrimaculatus, Anopheles sp., BLANK, Coquillettidia perturbans, Culex erraticus, Culex nigripalpus, Culex quinquefasciatus, Culex sp., Culiseta melanura, Mansonia genus sp., Mansonia titillans, Melanoconion Sp., Psorophora ciliata, Psorophora columbiae, Psorophora ferox, Psorophora sp., Toxorhynchites rutilus, Unknown unknown, Uranotaenia lowii, Uranotaenia sapphirina, Uranotaenia sp., Wyeomyia sp.

Years: 2016, 2017, 2018, 2019, 2010, 2011, 2012, 2013, 2014, 2015

Collection Method(s): BG-SENT bg-lure;co2, CDCLIGHT co2;light, NJLT octenol

Publication Title(s): 2010-2019 Collier County

Publication Description(s):

Location(s): Collier County, Florida, USA

Data Source

()

Imported by: Sam Rund

Dataset DOI: No DOI

Data

69739 total records in dataset

Species	Sample Start Date	Sample End Date	Sample Value	Sample Lat Dd	Sample Long Dd	Sampling Method
Uranotaenia sp.	2015-10-05	2015-10-05	16.0	26.4440703	-81.4314981	NJLT octenol
Melanoconion Sp.	2015-10-05	2015-10-05	64.0	26.4440703	-81.4314981	NJLT octenol
Mansonia genus sp.	2015-10-05	2015-10-05	8.0	26.4440703	-81.4314981	NJLT octenol
Culex nigripalpus	2015-10-05	2015-10-05	984.0	26.4440703	-81.4314981	NJLT octenol
Anopheles sp.	2015-10-05	2015-10-05	304.0	26.4440703	-81.4314981	NJLT octenol

Collections by Date

Number of Collections

Jan/ 1/2010 Jan/ 1/2012 Jan/ 1/2014 Jan/ 1/2016 Jan/ 1/2018

• Collections by Date

Population Abundance

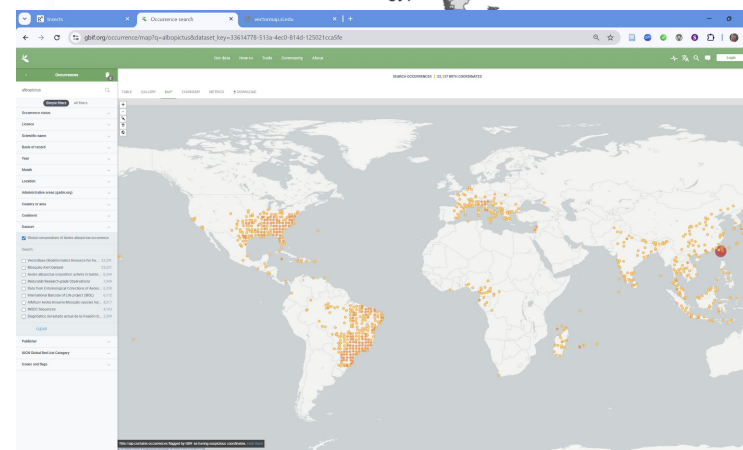
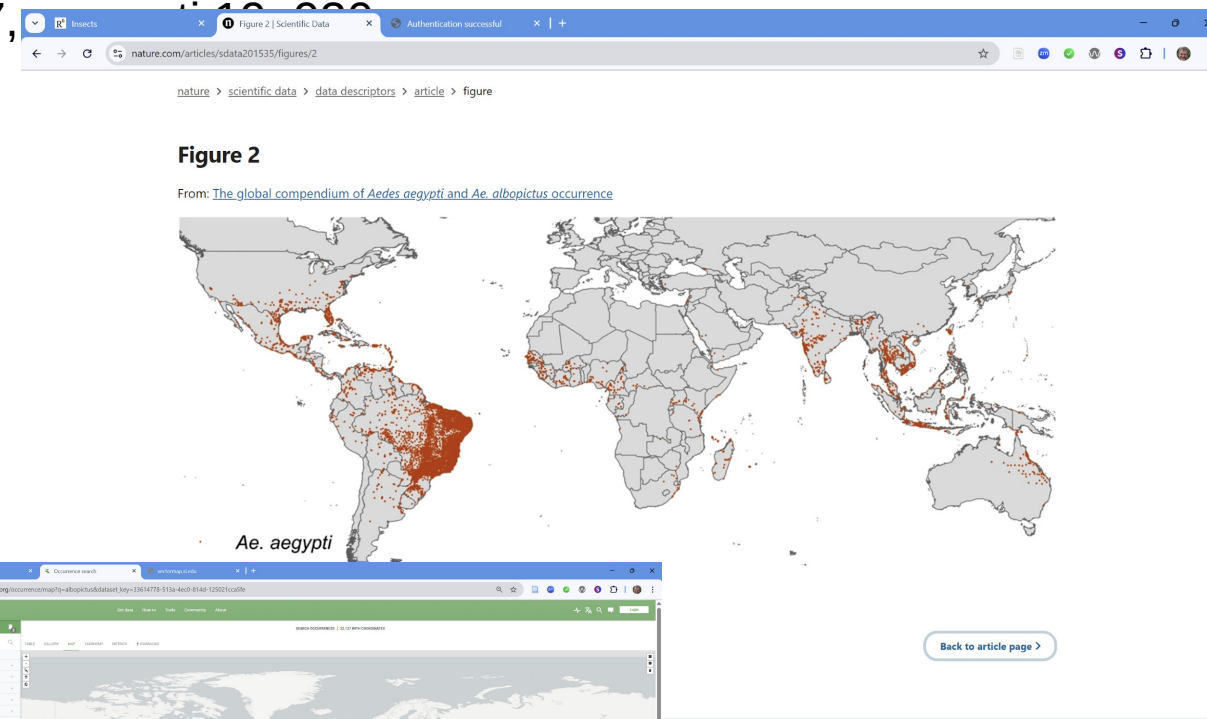
Zoom 1m 3m 6m YTD 1y All Feb 1, 2010 → Apr 26, 2019

Specimens Collected

Jan/ 1/2011 Jan/ 1/2012 Jan/ 1/2013 Jan/ 1/2014 Jan/ 1/2015 Jan/ 1/2016 Jan/ 1/2017 Jan/ 1/2018 Jan/ 1/2019

BLANK Aedes sp. Culex sp.

SOURCES: ONLINE Global compendium, Kraemer et al



The godfather of them all,
reformatted by Page et al, for

SOURCES:



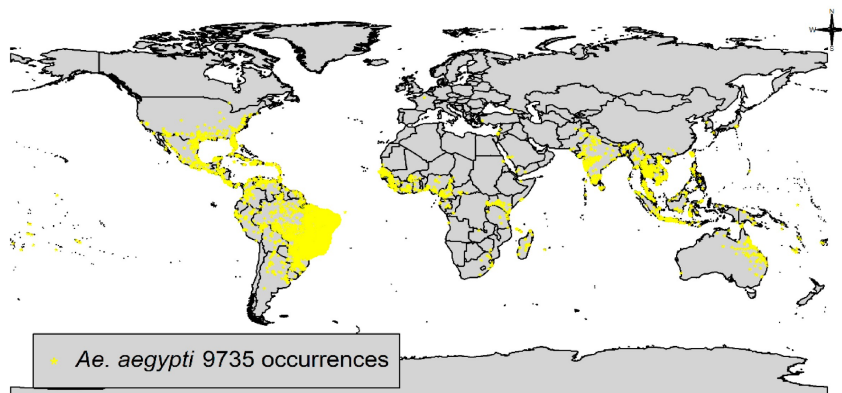
VectorMap: Laporta, G.Z.; Potter, A.M.; Olliaro, J.L.; Bourke, B.P.; Pecor, D.B.; Linton, Y.-M. Global Distribution of *Aedes aegypti* and *Aedes albopictus* in a Climate Change Scenario of Regional Rivalry. *Insects* **2023**, *14*, 49.

<https://doi.org/10.3390/insects14010049> (vectormap.si.edu) now at

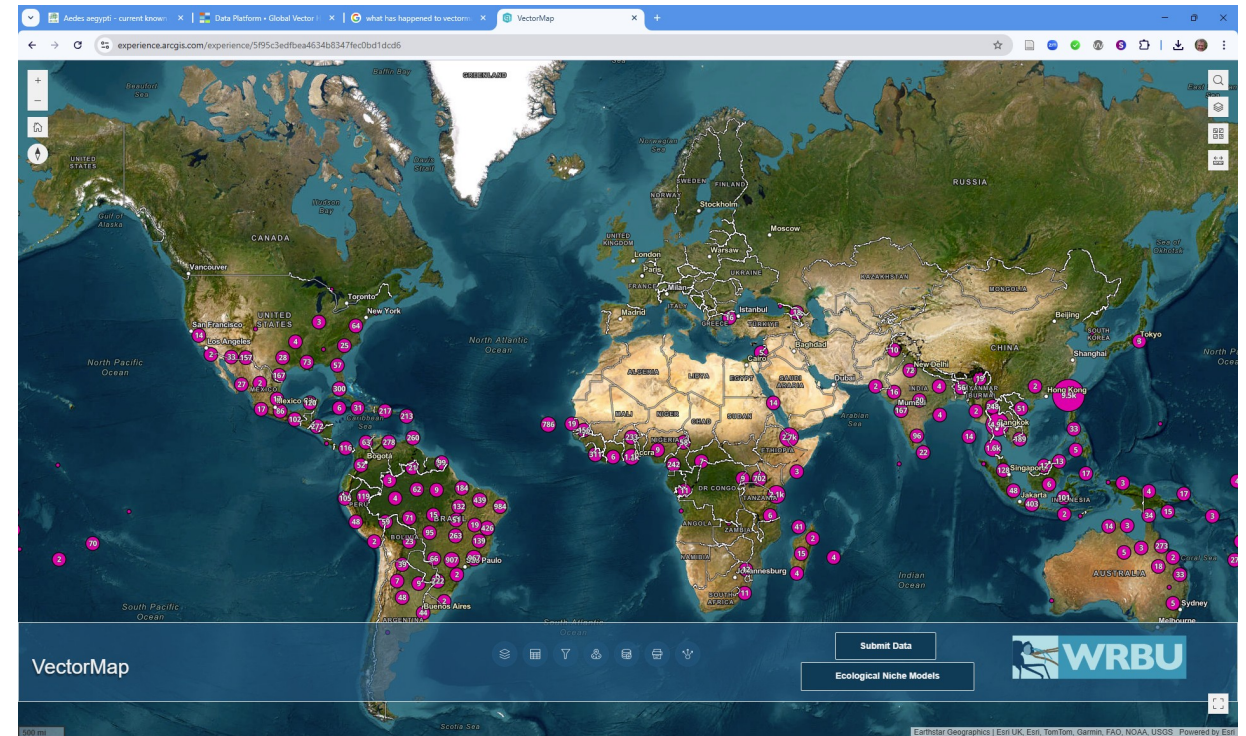
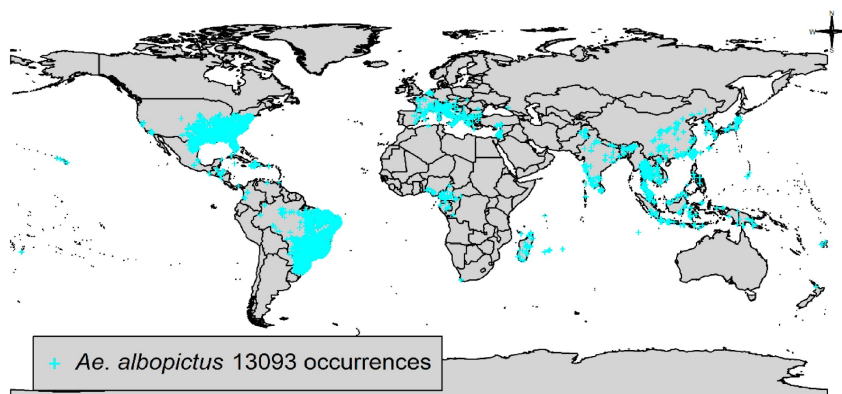
<https://experience.arcgis.com/experience/5f95c3edfbea4634b8347fec0bd1dcd6>

aegypti 38.3K, not sure about *albo pictus*, also hosts, bloodmeals, Insect Resistance

A



B

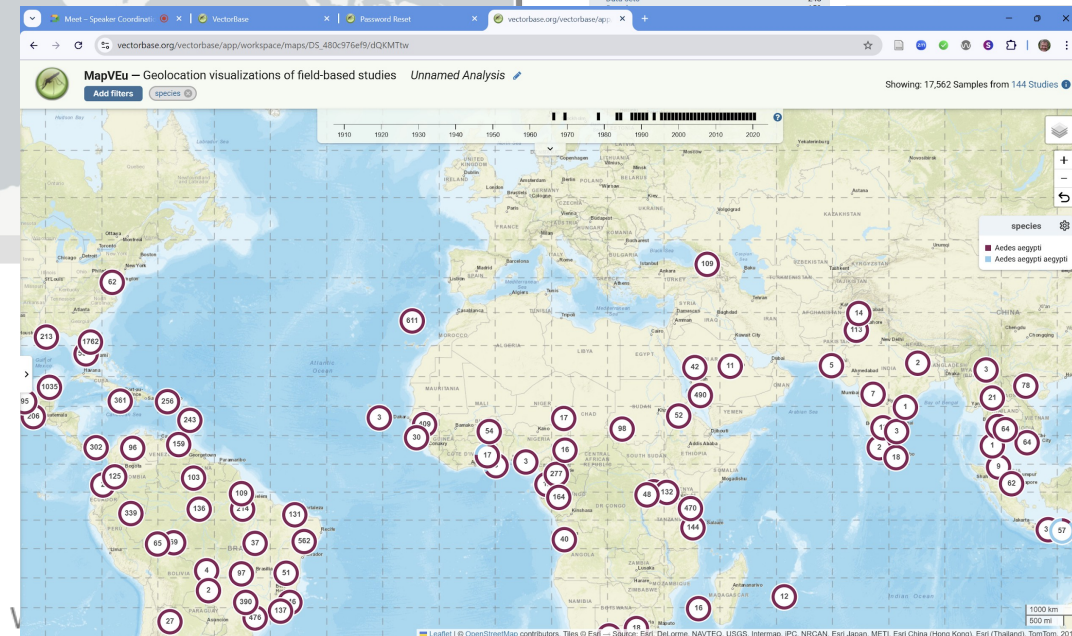
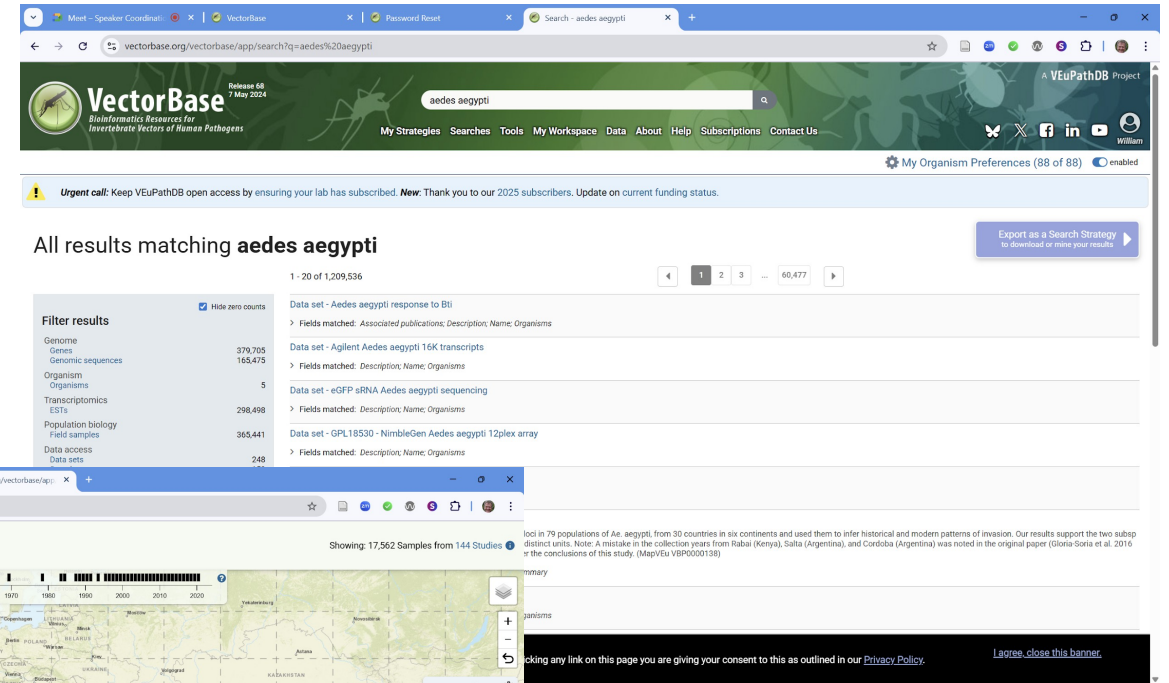
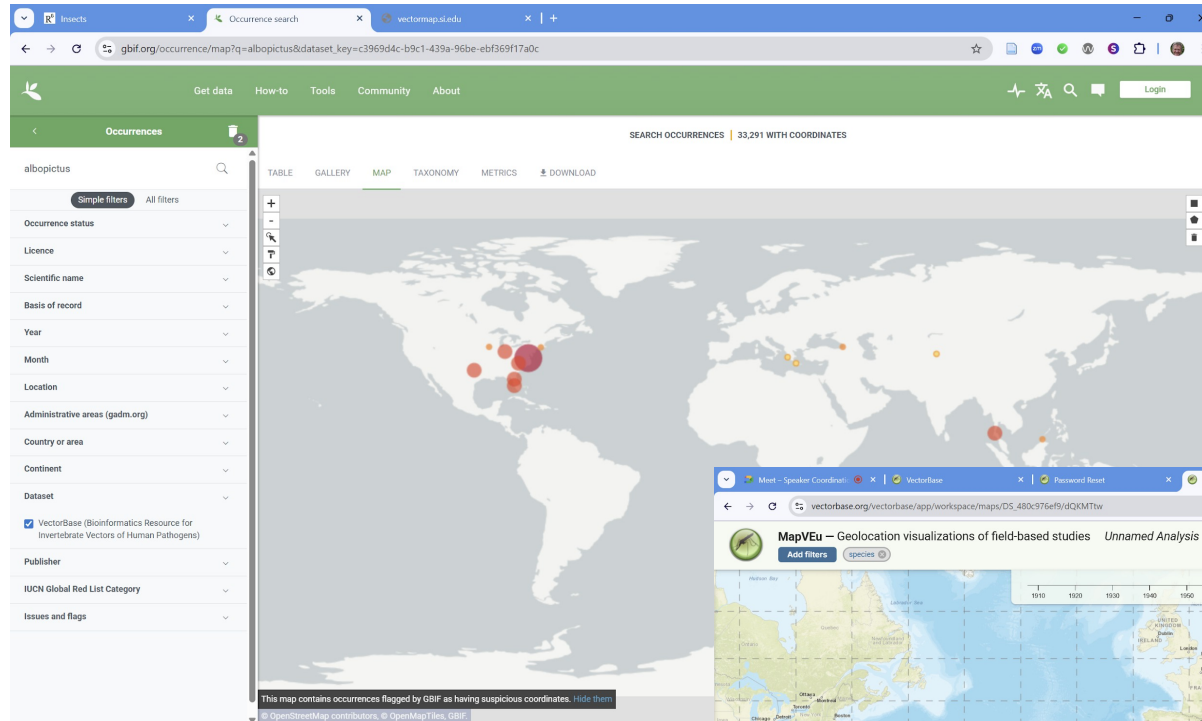


Failed to find any way to
download

William Wint, Available *Aedes* datasets, Modelling workshop, Cyprus, Sept 25

SOURCES:

ONLINE
VectorBase <https://vectorbase.org/vectorbase/app>.
1.6M records, 33,291 albopictus, 14,561 aegypti
Genome details , **NOW SUBSCRIPTION**

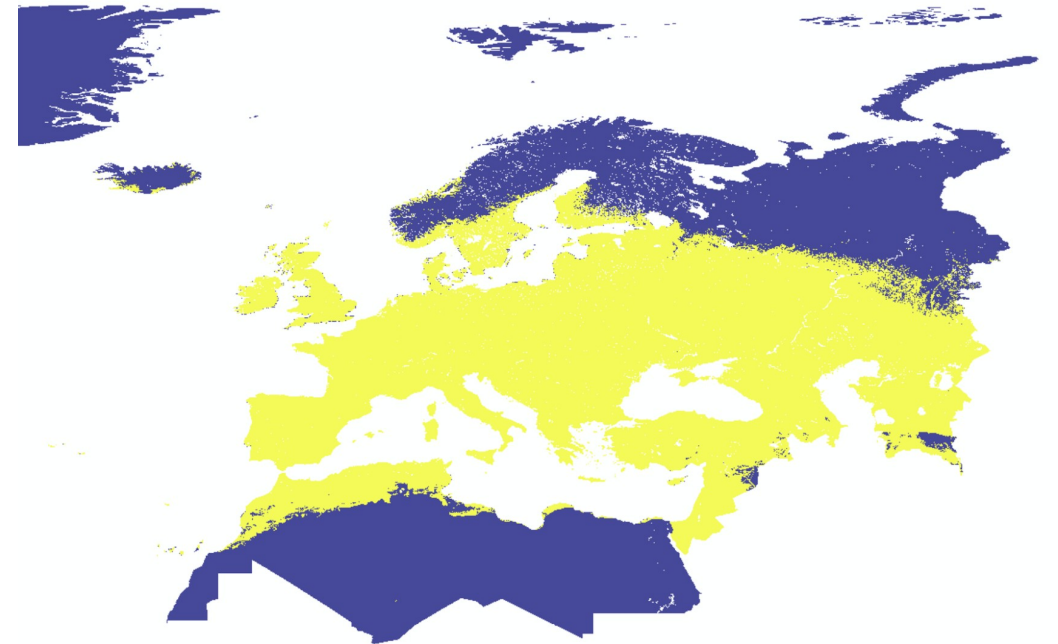
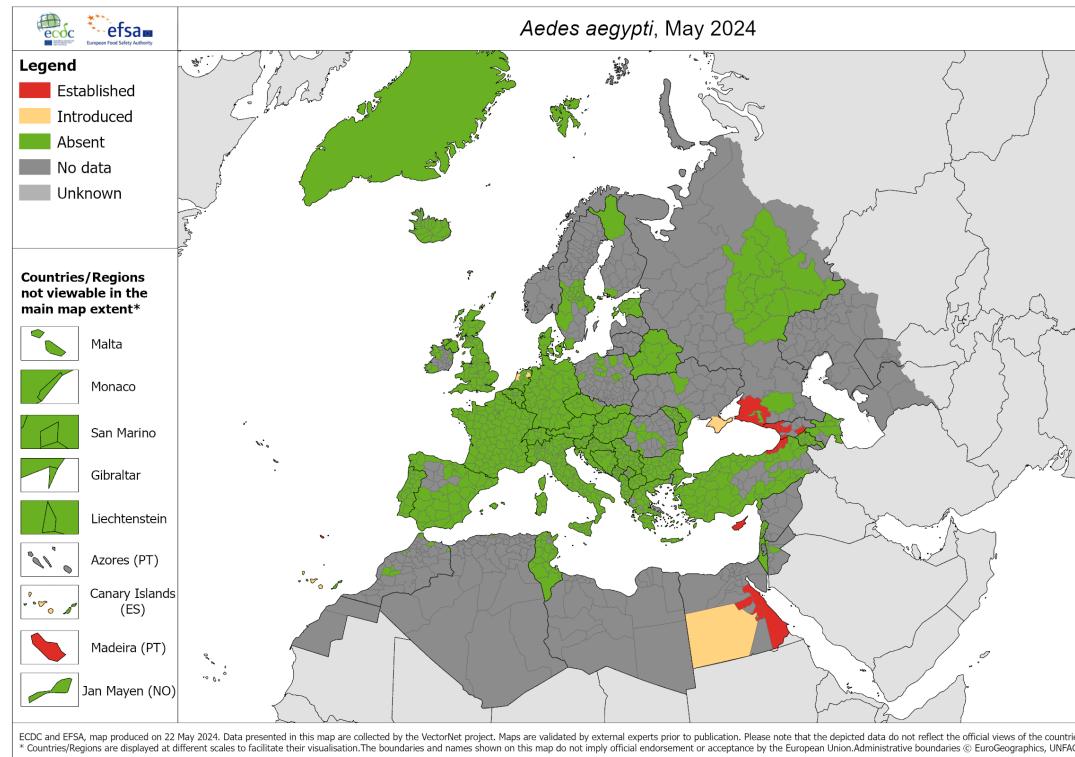


SOURCES: ONLINE



VectorNet

Mostly known for status maps
records with mixture of polygon, point, presence absence and abundance data. Literature and fieldwork. Also (soon) calculated absence from environmental thresholds and also from sampling methodology. To be added to GBIF. 40,239 also, limited number aegypti



Mosquito Alert

SOURCES: ONLINE



23K *albopictus* Europe centric no *aegypti* But rapid reaction. and more complete for



MOSQUITO
ALERT



SOURCES: ONLINE



Has sampling effort, burners, derived presence, many duplicates, need to be selective

albopictus 135K,

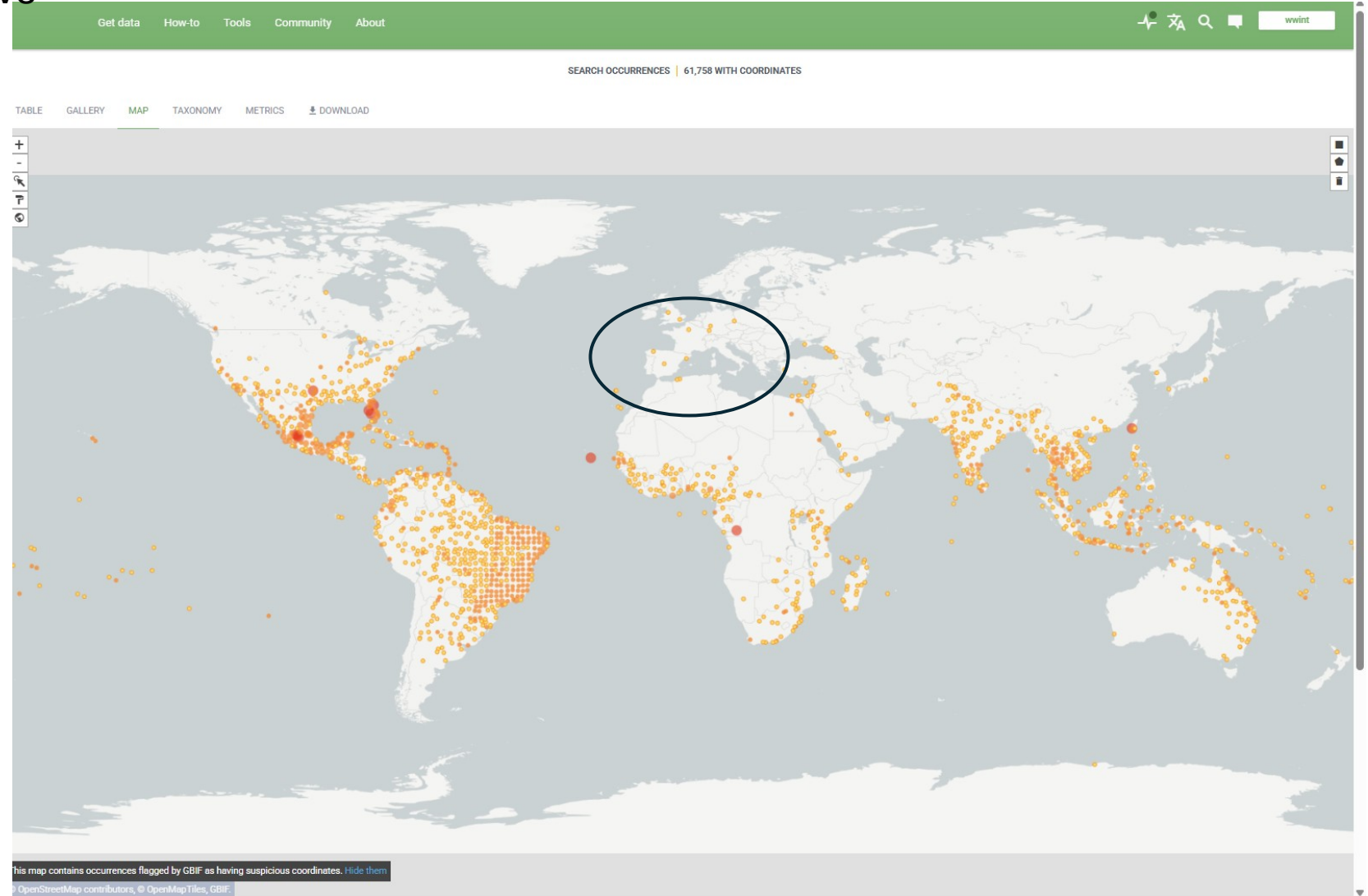
aegypti 63K

Most major datasets. Not yet VectorNet

Caveat emptor

Search

- ☐ Global compendium of *Aedes aegypti* occ... 19,929
- ☐ VectorBase (Bioinformatics Resource for I... 14,567
- ☐ Apoyo para la infraestructura de la colecció... 6,556
- ☐ Actualización de la Colección de Artrópodo... 6,215
- ☐ iNaturalist Research-grade Observations 3,617
- ☐ Fiocruz/CMN - Coleção de Mosquitos Neot... 3,095
- ☐ INSDC Sequences 3,012
- ☐ *Aedes* mosquito distribution across urban ... 1,939
- ☐ Collection of entomological, demographic, ... 1,920
- ☐ Naturalis Biodiversity Center (NL) - Diptera 1,502



DATA WRANGLING



Abundance equilibrations:

	Ovitrap	larval	BG traps		CDC traps		EVS traps		Human
		dips	unbaited	baited	Unbaited	baited	unbaited	baited	landing
Number Equivalents									
<i>. albopictus</i>	100		3	8	1	3	1	3	10
<i>. caspius</i>				10	5	100	1	40	50
<i>. japonicus</i>	15		1	2			1	2	3
<i>. vexans</i>			5	10	5	100	1	40	50
<i>. pipiens</i>		5	5	10		50		20	6
<i>thelleri</i>		5		10				20	
Conversion Factors									
<i>. albopictus</i>	1		33.3	12.75	100	33.3	100	33.3	10
<i>. caspius</i>				4	8	0.4	40	1	0.8
<i>. japonicus</i>			2	1			2	1	0.67
<i>. vexans</i>			8	4	8	0.4	40	1	0.8
<i>. pipiens</i>		4	4	2		0.4		1	3.3
<i>thelleri</i>		4		2				1	

Aggregation to de-duplicate:

Easy for for PA

1, 5 or 10km. 1 or 10 easiest as can use lat long decimal number
can be by date as well: annual abundance aggregated as Maximum

Less easy for abundance: standardised max (or ?mean) per sample unit per
date

Can be done by date/month

Absences for PA:

Not pseudo absence - unless certain that sampling is complete
should be via suitability thresholds – to allow absence near areas of presence